Srikrishnaa Vadivel

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Summary

Software engineer and Yale PhD researcher building scientific computing, high-performance simulation, data visualization, and ML infrastructure. Strong background in Python, C/C++, Linux, CUDA/OpenGL, MPI/OpenMP, Docker, Kubernetes, and production engineering for embedded and server-side systems.

Experience

Yale University, PhD Researcher 2021–present

- Built first-principles workflows for self-trapped excitons using HPC resources at Oak Ridge, Aurora, LBNL, and LLNL.
- Developed Python/C++ automation around PETSc/SLEPc, MPI, OpenMP, CMake, Linux shell tooling, containers, and parallel job execution.
- Prototyped ML models for molecular and materials data, including EGNN force models and diffusion-style generative workflows.

Probe Technologies / Weatherford, Software and Embedded Engineer 2019–2021

- Wrote ASP server and Microsoft database tooling for field data systems.
- Built CUDA/OpenGL acoustic logging visualization software for processing and displaying large waveform data sets.
- Delivered embedded firmware for C8051-class MCUs and Lattice FPGA sensor logging used in geological well tools; validated with oscilloscopes and field tests.

FindLight, Photonics Blogger 2018–2019

- Wrote technical photonics articles translating optics products and applications into clear engineering content.

Projects

- **DOLFINx PDE Gallery**: weak-form derivations, Poisson solver, PyVista visualization, and docs in index notation.
- **Meep Ring Resonator**: FDTD photonics simulation of a ring coupled to a waveguide with field animation workflow.
- **First-Principles LiF**: plane-wave Hamiltonian scaffold and band plotting for DFT-style workflows.
- **ROS 2 TurtleBot Simulation**: Docker-oriented Nav2/Gazebo plan with trajectory analysis fallback.

Education

Yale University, PhD, Electrical Engineering 2021–present
Cornell University, MEng, Electrical and Computer Engineering 2017–2018
Cornell University, BA, Electrical and Computer Engineering 2013–2017

Skills

Python, C, C++, CUDA, OpenGL, Bash, Linux, CMake, PETSc, SLEPc, MPI, OpenMP, Docker, Kubernetes, SQL Server, ASP, PyTorch, NumPy, SciPy, pandas, scikit-learn, Hugging Face, Terraform, AWS, GCP.